

Lecture 1: Neural Network Fundamentals

A neural network is composed of layers of interconnected nodes, or neurons, each applying a weighted sum followed by a non-linear activation function such as ReLU, Sigmoid, or Tanh.

Backpropagation

Backpropagation is the algorithm used to compute gradients of the loss function with respect to each weight by applying the chain rule, allowing the network to update its parameters via gradient descent.